

Learning-Rate Schedules Are Not Adiabatic: A Rate-Dependent Correction for Loss-Curve Prediction

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Peking University Deep Learning Course Task 2
github.com/wjjpku/DL-final

Self-contained summary

MPL and FSL are strong *adiabatic* predictors: they model the quasi-static loss implied by the LR history. This project identifies the missing transient: when LR is annealed faster than the optimizer's excess-loss state can relax, the true loss remains **above** the quasi-static prediction. I derive a low-capacity correction, test it on public LLM curves, and reproduce the mechanism in a real $\sim 10\text{M}$ transformer and controlled AdamW simulations.

Standalone report: project goal, method, experimental settings, results, analysis, limitations, and next steps.

Roadmap

- ① Project Goal
- ② Data and Reproduction
- ③ Method
- ④ Experiments on Public LLM Curves
- ⑤ Independent Reproduction
- ⑥ Synthesis

Task 2 coverage checklist

This deck is organized to explicitly cover the six required components.

Requirement	Where it appears in this deck
Problem background and goal	Project Goal section: schedule-aware loss-curve prediction, motivation, and the core question.
Data processing and experimental setup	Data and Reproduction: data source, preprocessing pipeline, splits, fitting protocol, and parameters.
Reproduction results	Reproduction slide: Tissue/MPL reproduction and public-curve baseline validation.
Method and result comparison	Public-curve experiments: MPL vs MPL+DropRelaxS, transfer table, and fair-baseline diagnostic.
Analysis and reflection	Independent reproduction, negative results, limitations, and next-work slides.
Code and work division	Reproducibility map plus explicit code/running/division slide.

Design choice. The slides are not a speaking outline; they are written so that the project can be understood by reading the deck alone.

The problem: predict the whole loss curve before training

Goal. Given a learning-rate schedule $\{\eta_t\}$, predict the pretraining loss curve $L(t)$ without running the full training job.

Why this matters.

- LLM pretraining is expensive; schedule search by full trial runs is wasteful.
- Standard scaling laws mostly target endpoint loss; schedule-aware laws must predict the *trajectory* under warmup, stable phases, cooldowns, and WSD-style drops.
- The hardest test is cross-schedule transfer: fit on slow schedules such as cosine, then predict unseen sharp-decay WSD curves.

Project question.

Do current loss-curve laws only know how much LR was spent, or also how fast it was spent?

Baseline laws: strong, but structurally adiabatic

Let $S(t) = \sum_{i \leq t} \eta_i$ be cumulative LR and $\Delta\eta_k = \eta_{k-1} - \eta_k$ be a LR decrement.

Multi-Power Law (MPL).

$$L(t) = L_0 + A S(t)^{-\alpha} + B \sum_{k \leq t} \Delta\eta_k G(\eta_k^{-\gamma}(S(t) - S(k))), \quad G(x) = 1 - (1 + Cx)^{-\beta}.$$

Functional Scaling Law (FSL). FSL gives a closely related functional predictor over the LR history.

What they capture well.

- Integrated progress through $S(t)$.
- A saturating benefit from past LR drops.
- Cross-schedule structure in public LLM curves.

The gap. Their response to a drop is essentially quasi-static: after fitting the schedule kernel, they do not explicitly model the finite-time tracking error of a moving LR-dependent noise floor. I call this an **adiabatic approximation**.

**Loss depends not only on how much LR has been spent,
but also on how fast it was spent.**

Mechanism.

- At a fixed LR, the optimizer has an equilibrium excess-loss/noise-floor state.
- A sharp LR drop lowers that equilibrium immediately.
- The actual excess-loss state relaxes only over a finite time.
- During this transient, the true loss is **higher** than the adiabatic prediction.

Resulting signature. Slow and fast schedules may reach the same final LR, but the fast sweep retains a positive residual. This is not just a function of cumulative LR S ; it is a rate-dependent lag.

Data source and processing pipeline

Public-curve data.

- Source: official MPL public loss curves vendored under `external/MultiPowerLaw/loss_curve_repo/csv_{25,100,400}`.
- Scales: 25M, 100M, 400M.
- Schedules: cosine, constant, WSD, WSD-long-decay, and WSD constant-tail probes.

Processing pipeline.

- 1 Load per-run CSV curves and learning-rate schedules.
- 2 Align curves by training step; compute cumulative LR $S(t) = \sum_{i \leq t} \eta_i$ and LR decrements $\Delta\eta_k = (\eta_{k-1} - \eta_k)_+$.
- 3 Fit baseline laws on selected training curves using log-loss Huber objectives.
- 4 Evaluate predictions on held-out WSD-family curves using MAE.
- 5 Compute residuals $L_{\text{true}} - L_{\text{MPL}}$ and fit/test DropRelaxS only on the residual structure.

Main parameters. MPL has $(L_0, A, \alpha, B, C, \beta, \gamma)$; DropRelaxS adds λ_{slow} , c , and independently measured $\chi = dL_{\text{eq}}/d\eta$.

Experimental split and evaluation protocol

Course-task reproduction split.

- Training curves: `cosine_24000`, `cosine_72000`.
- Held-out test curves: `wsd_20000_24000`, `wsd1d_20000_24000`, `wsdcon_3`, `wsdcon_9`, `wsdcon_18`.
- Evaluation scales: 25M, 100M, and 400M.

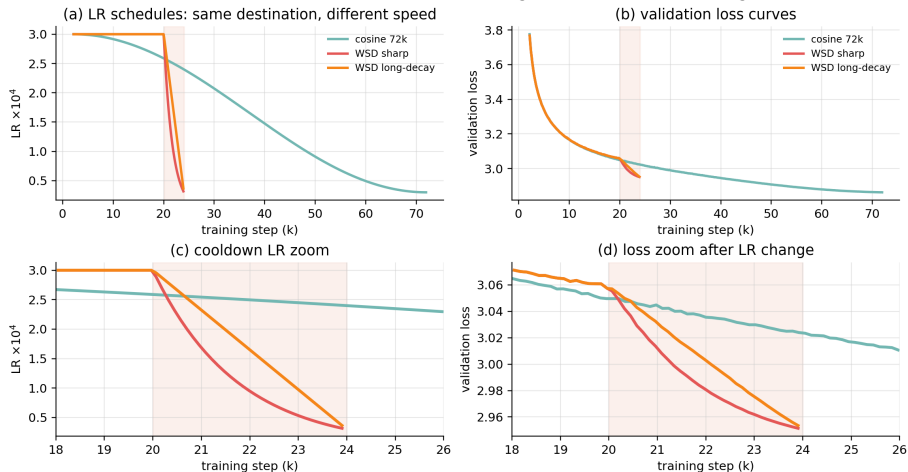
Additional public-curve experiments.

- Use a faithfully reproduced MPL as the adiabatic baseline.
- Measure τ from two-stage `wsdcon` constant-tail windows.
- Use `wsd/wsdl1d` as held-out sharp-decay targets for low-calibration cross-scale transfer.

Why this protocol is fair. The correction is evaluated after MPL has already captured the backbone and schedule-drop structure. Therefore any remaining positive residual is not simply a weak baseline artifact.

Data visualization: WSD isolates LR-change speed

Data focus: WSD tests whether fast LR change leaves a transient loss lag

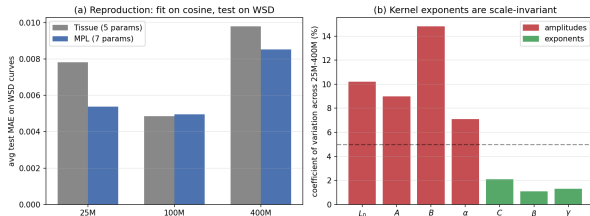


The key contrast is local: WSD reaches low LR much faster, so the cooldown window is where rate dependence should appear.

Reproducing existing methods: Tissue and MPL

Scale	Tissue	MPL
25M	0.00783	0.00539
100M	0.00486	0.00496
400M	0.00979	0.00852

Average WSD-family test MAE.



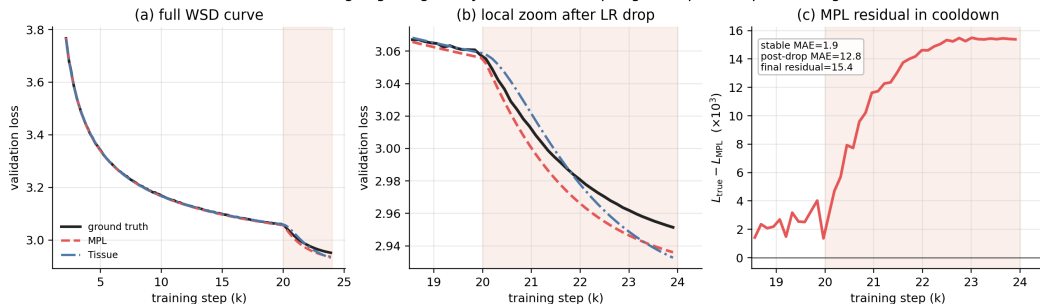
Key differences and difficulties.

- Tissue is compact and interpretable but less stable on sharp WSD families.
- MPL is higher-capacity and matches the published public-curve results.
- Correct optimizer protocol matters: weak or inconsistent fitting can make false “improvements” look real.

Conclusion. The reproduced MPL is the baseline used for all residual analysis.

Curve-level fitting: zoom into the WSD cooldown

100M WSD: fitting is good globally, but the LR-drop region exposes a positive lag



Globally the fit is good; locally after the LR drop, MPL predicts too low, leaving a positive residual.

Theory in one slide: linear response of a moving equilibrium

Assume an LR-dependent hidden excess-loss state z_t with local equilibrium $z^*(S_t, \eta_t)$:

$$z_{t+1} = \Phi_{S_t, \eta_t}(z_t), \quad L_t = L_{\text{ad}}(S_t, \eta_t) + r_t.$$

Linearizing around the moving equilibrium gives the cooldown-specific residual

$$r_t \approx \sum_{k < t} K_{k,t} (\eta_k - \eta_{k+1})_+.$$

Under the first-order optimizer approximation $T_t \simeq I - \eta_t A$, products of transition matrices become exponentials in **cumulative-LR time**:

$$T_{t-1} \cdots T_{k+1} \simeq \exp[-A(S_t - S_k)].$$

Primitive prediction

The non-adiabatic residual is a causal convolution of LR decrements with a stable memory kernel in S-time, not a free bump fitted after the fact.

Operational correction: DropRelaxS

The full kernel can be a spectral mixture. To avoid overfitting, the paper uses the lowest-capacity one-pole closure:

$$L(t) = L_{\text{MPL}}(t) + c \chi \sum_{k < t} e^{-\lambda_{\text{slow}}(S_t - S_k)} (\eta_k - \eta_{k+1})_+$$

$$\chi = \frac{dL_{\text{eq}}}{d\eta}.$$

Interpretation.

- $(\eta_k - \eta_{k+1})_+$: only LR drops create the positive lag.
- $e^{-\lambda_{\text{slow}}(S_t - S_k)}$: memory decays in cumulative LR time.
- χ : amplitude equals the equilibrium noise-floor sensitivity.
- c : low-capacity compression of a spectral mixture into one pole.

Important convention. If drops are normalized by η_{peak} , the prefactor must absorb that factor. With raw LR drops, the dimensionally clean amplitude is $c\chi$.

AdamW specialization: expected relaxation scaling

For a noisy quadratic mode with curvature λ_i and gradient-noise standard deviation s_i , noise-dominated AdamW has $v_i^* \simeq s_i^2$. Locally it behaves like preconditioned SGD:

$$x_{i,t+1} = \left(1 - \eta_t \frac{\lambda_i}{s_i}\right) x_{i,t} - \eta_t \frac{1}{s_i} \xi_{i,t}.$$

For $V_i = \mathbb{E}[x_i^2]$:

$$V_i^*(\eta) = \frac{\eta s_i}{2\lambda_i} + O(\eta^2), \quad F_i^*(\eta) = \frac{1}{2} \lambda_i V_i^*(\eta) = \frac{\eta s_i}{4} + O(\eta^2).$$

Two useful consistency checks.

- **Amplitude identity:** $\sum_i dF_i^*/d\eta = \frac{1}{4} \sum_i s_i + O(\eta)$.
- **Classical relaxation time:**

$$\tau_i^{-1} = 2\eta \frac{\lambda_i}{s_i}, \quad \tau_i \propto \eta^{-1}.$$

This timescale is not a new theorem; here it is used as a mechanistic check for whether the **residual left by adiabatic loss-curve laws** behaves like finite-time relaxation.

Predictions and falsifiability

The correction predicts six observable signatures.

- 1 Residual after fast LR drops is positive.
- 2 Residual depends causally on LR decrements, not just on LR level.
- 3 Memory is in cumulative LR time S , not raw step time.
- 4 Amplitude is controlled by $dL_{\text{eq}}/d\eta$.
- 5 In a noise-dominated AdamW regime, residual relaxation should be consistent with $\tau \propto 1/\eta$.
- 6 A spectral mixture should sometimes beat a single exponential.

Failure modes are also meaningful.

- If a full set of WSD curves is available, MPL can absorb the residual by refitting.
- If the scale is too small/noisy, this relaxation scaling may not be resolved.
- A single decay curve can make finite- λ_{slow} nearly degenerate with the adiabatic limit; rate dependence must be identified across schedules.

Public-curve experimental setting

Data.

- Official MPL public LLM loss curves.
- Llama-2-style models at 25M, 100M, and 400M.
- Nine schedules per scale, including cosine, constant, WSD, WSD-long-decay, and two-stage WSD constant-tail probes.

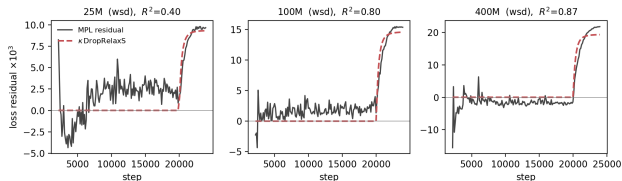
Baseline.

- Use a faithfully reproduced well-fit MPL.
- Test MAE matches Luo et al.: 0.0054/0.0050/0.0085 at 25/100/400M.
- This matters: the residual is measured after a strong adiabatic baseline, not after a weak fit.

Evaluation.

- Residual structure on held-out sharp decays.
- τ measured from two-stage `wsdcon` relaxation transients.
- Cross-scale transfer on held-out `wsd`/`wsdl1d` curves.

Result 1: fast decays leave a structured positive residual



Observation.

- During the stable phase, the well-fit MPL residual is near zero.
- At a sharp WSD cooldown, the residual jumps positive.
- The one-pole DropRelaxS term tracks the residual.

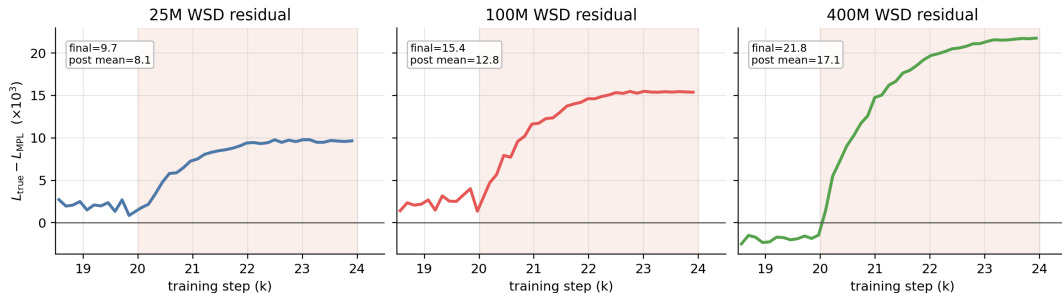
Fit quality.

$$R^2 = 0.40/0.80/0.87 \quad (25/100/400M).$$

Interpretation. The signal strengthens with scale. The residual is exactly the expected lag: after a fast drop, the true loss stays above the quasi-static prediction.

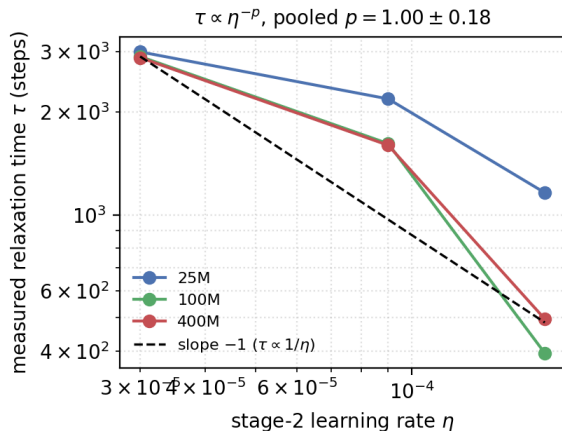
Result 1b: WSD cooldown residual across scales

The WSD cooldown residual is positive across public model scales



The same post-drop sign appears at 25/100/400M, and the residual magnitude is larger at larger scale.

Consistency check: relaxation time vs. LR



Protocol.

- Use two-stage wsdcon curves.
- At step 8000, LR drops to a constant second-stage value: $\eta \in \{3, 9, 18\} \times 10^{-5}$.
- Fit the residual transient $r(t) \propto e^{-(t-8000)/\tau}$.

Observed trend.

$$\tau \propto \eta^{-p}, \quad p = 1.00 \pm 0.18$$

pooled over public curves; 100/400M track the classical slope -1 most cleanly.

Interpretation. This supports the finite-time relaxation mechanism, but it is not the main claim: the main claim is the positive WSD residual and its useful correction.

Result 3: amplitude is tied to the noise floor

Independent amplitude probe. Use final losses of two-stage constant-tail curves to estimate the equilibrium noise-floor sensitivity $dL_{\text{eq}}/d\eta$ without fitting DropRelaxS to the target curve.

Scale	25M	100M	400M
$\kappa_{\text{fit}}/\kappa_{\text{pred}}$	0.43	0.51	0.60

Analysis.

- The ratio has coefficient of variation 14% across scales.
- The < 1 factor is expected: the single-pole closure compresses a spectral mixture.
- With raw LR drops this gives $c \approx 0.5$; with normalized drops $(\eta_k - \eta_{k+1})_+/\eta_{\text{peak}}$, the same convention requires $c \approx 60\text{--}86$.

Practical conclusion

The transferable object is the near-constant ratio under a clearly stated normalization convention, not a universal first-principles value of c .

Result 4: low-calibration cross-scale transfer improves sharp decays

Transfer protocol.

- Fix $\lambda_{\text{slow}} = 10$ from the public-curve relaxation measurement.
- Take c from a leave-one-scale-out average of the other scales.
- Measure $dL_{\text{eq}}/d\eta$ from cheap two-stage probes at the target scale.
- Do **zero fitting on the held-out target curve**.

Scale	wsd		wsdld		avg. change
	MPL	MPL+DropRelaxS	MPL	MPL+DropRelaxS	
25M	0.00341	0.00246	0.00314	0.00219	−29%
100M	0.00375	0.00164	0.00325	0.00161	−53%
400M	0.00470	0.00236	0.00385	0.00212	−47%
overall					−44% (6/6 wins)

Analysis. The correction is most useful in the data-poor regime: cosine plus cheap noise-floor probes, but no full WSD curves.

Fair-baseline diagnostic: formula gap vs information gap

Key question. Is DropRelaxS a better formula, or does it just supply information missing from a few-shot MPL fit?

Diagnostic.

- If MPL is refit on several full WSD curves, it can absorb the non-adiabatic residual.
- In that data-rich setting, the explicit DropRelaxS term adds little.
- In the data-poor setting, DropRelaxS provides a structured prior that transfers from cheap probes and other scales.

Honest scope

The correction is an **information-efficient leading-order term**. It is not meant to beat a fully refit, data-rich MPL on its own training distribution.

Calibration audit: staying within the current law

Question. Can the current law be made more reliable by changing only how λ_{slow} and the amplitude are calibrated?

Fixed model class.

$$L(t) = L_{\text{MPL}}(t) + \kappa \text{DropRelaxS}_\lambda(t), \quad \text{DropRelaxS}_\lambda(t) = \sum_{k < t} e^{-\lambda(S_t - S_k)} (\eta_k - \eta_{k+1})_+.$$

Calibration protocol	Best λ	Final WSD MAE change	wins
fit κ on <code>wsdcon_9</code> probe	10	−14.5%	6/6
fit κ on all <code>wsdcon</code> probes	7–10	−17.9%	6/6
cross-scale c + noise-floor sensitivity	5–10	−44.0% to −45.0%	6/6

Interpretation.

- The current law improves final cosine-fit MPL $\rightarrow$ WSD prediction under several non-target calibration choices.
- $\lambda = 10$ is not a hand-tuned optimum; it is close to the best range and is the independently measured value from the relaxation probe.
- The large 44% gain relies on the stronger cross-scale amplitude chain, not merely on fitting residuals on one probe curve.

Final kappa estimator: remove nuisance drift before transfer

Problem with raw amplitude fitting. The residual after MPL contains both the schedule-response feature and smooth MPL drift. Directly fitting κ can get the trend direction right but the magnitude wrong.

Final calibration model.

$$r = L_{\text{obs}} - L_{\text{MPL}} = \kappa_{\star} \phi + g_{\star} + \varepsilon, \quad g_{\star} \in \mathcal{G}, \quad \kappa_{\star} \geq 0.$$

Let $M_{\mathcal{G}}$ residualize against the low-frequency nuisance subspace:

$$\phi_{\perp} = M_{\mathcal{G}} \phi, \quad r_{\perp} = M_{\mathcal{G}} r, \quad R = \frac{\|\phi_{\perp}\|_2^2}{\|\phi\|_2^2}.$$

Cap-free paper-facing estimator.

$$\hat{\kappa} = \sqrt{R} \left(\frac{\langle \phi_{\perp}, r_{\perp} \rangle}{\|\phi_{\perp}\|_2^2 + \tau^2} \right)_{+}, \quad \tau = \sigma/k_0.$$

Interpretation: FWL partial regression + empirical-Bayes ridge + identifiable-amplitude conversion.

Final kappa audit: what changed

Cross-schedule transfer matrix.

Estimator	worst offdiag	mean offdiag	cosine→WSD	max cosine κ
smooth baseline	−0.0%	−10.1%	−0.0%	0.0000
EB without nuisance control	−1.0%	−10.8%	−3.1%	0.0045
final cap-free	−2.7%	−12.1%	−4.3%	0.0089
spectral G_4 audit	−1.8%	−10.0%	−3.6%	0.0070
no identifiable conversion	+1.8%	−12.2%	−17.4%	0.0300

Readout.

- The DCT low-frequency audit is slightly more conservative, but shows the result is not tied to a polynomial-shaped nuisance basis.
- The hard cap is not the mechanism: cap-free and capped legacy variants have the same worst off-diagonal result.
- Removing $\sqrt{R}$ creates a positive off-diagonal failure.
- Train-only τ preserves the conclusion: worst remains −2.7%, mean remains −12.1%, and cosine→WSD becomes −5.6%.

Scripts: `repro/current_law_final_kappa.py`, `repro/current_law_trainonly_tau_audit.py`.

How to choose the nuisance subspace

The object is low-frequency drift control, not a polynomial fit. Use a DCT low-frequency family:

$$\mathcal{G}_K = \text{span}\{q_0, \dots, q_K\}, \quad R_K = \frac{\|M_{\mathcal{G}_K} \phi\|_2^2}{\|\phi\|_2^2}.$$

Automatic spectral rule.

$$K^* = \arg \min_{K \geq K_{\min}} |R_K - \rho|.$$

Choice of $\mathcal{G}$	worst offdiag	mean offdiag	cosine→WSD
fixed DCT $K = 4$	-1.8%	-10.0%	-3.6%
retention target only	+304.0%	+8.4%	-9.5%
$K_{\min} = 3, \rho = 0.35$	-1.7%	-11.2%	-10.1%

Conclusion. First enforce minimum low-frequency drift control; then use identifiable feature energy to choose among sufficiently rich nuisance spaces.

Split audit: where the current law fails

Research question. If we calibrate λ and κ on arbitrary non-cosine curves, does the same law generalize to the rest?

Audit	Result	wins	Interpretation
all non-cosine train/test splits	median -14.1% MAE	182/225	usually helpful
probe-only $\rightarrow$ final <code>wsd/wsdld</code>	mean -14.8% MAE	42/42	robust target use case
sharp decays $\rightarrow$ <code>wsdcon</code> tails	up to $+96\%$ MAE	fail	over-corrects long tails

Conclusion. DropRelaxS should not be sold as a universal residual smoother. Its defensible role is narrower: a leading-order sharp-cooldown correction whose amplitude is best fixed by probe curves and cross-scale transfer, then judged on cosine-fit MPL $\rightarrow$ WSD prediction.

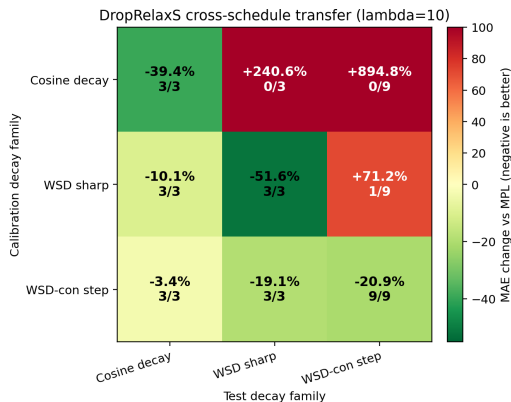
More calibration data: matched beats mixed

Question. If we allow more calibration curves, does the same law become more reliable?

Calibration data	λ choice	Held-out sharp MAE change	wins
other sharp curve only	fixed 10	−49.0%	6/6
other sharp curve only	selected on calibration	−49.9%	6/6
all <code>wsdcon</code> probes + other sharp	fixed 10	−19.5%	6/6
all <code>wsdcon</code> probes only	fixed 10	−17.6%	6/6

Research conclusion. More data is not automatically better. A full sharp-decay calibration curve transfers strongly to the other sharp shape, but mixing it with long `wsdcon` tails dilutes the amplitude. For the strict no-target-WSD setting, the cross-scale probe rule remains the cleaner claim.

Cross-schedule fitting matrix



Reading. Cosine calibration does not transfer to WSD targets; `wdcon` step probes transfer to sharp WSD with smaller but stable gain; sharp-WSD calibration over-corrects `wdcon` tails. The law is schedule-family dependent, not universal.

Method comparison: what each model can and cannot express

Tissue. Simple LR-integral baseline with few parameters. It is useful as a lightweight reference, but has a coarse schedule response and weaker WSD transfer.

MPL. Strong adiabatic schedule-aware predictor. Public results are reproducible, but lag is absorbed only when full WSD curves are available.

MPL+DropRelaxS. Adds a causal finite-time lag term with low calibration cost. It is most useful for data-poor sharp-decay prediction; constants are still measured and the single-curve degeneracy remains.

Empirical comparison.

- On the course split, MPL reproduces or improves the existing baselines at 25/400M and ties Tissue at 100M.
- On held-out sharp decays, low-calibration MPL+DropRelaxS reduces MAE by 44% over well-fit MPL in the public same-architecture setting.
- When full WSD curves are available, refit MPL absorbs much of the residual; the gain of DropRelaxS is specifically in the “predict before training” regime.
- Exhaustive split audits show the current law can over-correct `wsdcon` tails when calibrated only on sharp decays, so its scope is sharp-cooldown prediction, not arbitrary residual smoothing.

Why an independent reproduction was needed

Problem. The original project data are not fully recoverable: the author-side public loss-curve repository is no longer hosted.

Reproduction strategy.

- ① Build a controlled noisy-quadratic AdamW simulator from scratch.
- ② Train an independent $\sim 10\text{M}$ byte-level transformer on WikiText.
- ③ Run adversarial audits: check for estimator failures, degeneracies, leakage, and windowing artifacts.

Purpose. The reproduction is not just a rerun. It tests whether the mechanism survives outside the original public curves and clarifies which quantitative claims do *not* transfer at small scale.

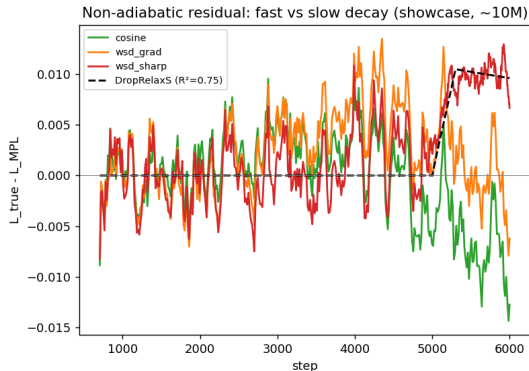
Real $\sim 10\text{M}$ transformer: no-fitting evidence of rate dependence

Schedule	final loss	cumulative LR S
cosine (gradual)	1.0490	8.20
wsd (gradual)	1.0482	9.22
wsd (sharp)	1.0593	12.59

All schedules reach the same final LR.

Analysis.

- Sharp sweep: higher loss despite 37% more cumulative LR.
- Any law monotone in S predicts the opposite.
- Across schedules, finite- λ_{slow} DropRelaxS beats the adiabatic limit: $\Delta R^2 \approx +0.18$.



Controlled AdamW simulation: quantitative mechanism checks

Simulator. Noisy quadratic model with real AdamW updates: true m and v EMAs, true $m/(\sqrt{v} + \epsilon)$ preconditioning, Gaussian gradient noise, and Monte-Carlo ground truth. Closed-form linear theory is used only as an independent baseline.

Claim tested	Result	Interpretation
τ vs LR	$p = 0.971$, log-log $R^2 = 0.9996$	consistent noise-dominated scaling
τ vs β_2	CV = 1.1% over $[0.9, 0.9999]$	lag not an EMA artifact
Amplitude identity	3.075 vs closed-form 3.0	within 3%
Spectral mixture	recover rates $\{0.87, 16.1\}$ for true $\{1, 16\}$	two-pole structure real

Falsifiability. A signal-dominated spectrum breaks the slope-1 pattern, so the check is not a circular fit to a fixed exponent.

What does not transfer at $\sim 10M$

Negative results are part of the final claim.

- The real $\sim 10M$ transformer does **not** resolve $\tau \propto 1/\eta$: over a $16\times$ LR range, measured post-step relaxation is roughly flat ($\tau \approx 475\text{--}912$ steps, $p \approx 0.18$).
- Audits reject the simple “window truncation” excuse: synthetic recovery tests and profile-likelihood checks reject very large hidden τ values.
- On a single decay curve, finite- λ_{slow} DropRelaxS is numerically close to the adiabatic limit (correlation ≈ 0.99). Rate dependence is identifiable only across schedules.
- Frozen cross-shape transfer in controlled simulations fails its pre-registered threshold (mean $R^2 = 0.52 < 0.6$).

Bounded conclusion

The qualitative non-adiabatic lag is robust. The inverse-LR relaxation trend is useful supporting evidence in public larger curves and controlled AdamW, but it is not established by the small real transformer.

What is solid

Main evidence.

- Fast LR decays create a positive residual relative to well-fit adiabatic laws.
- The residual has the predicted causal LR-decrement structure.
- Public LLM curves show residual relaxation consistent with inverse-LR scaling ($p = 1.00 \pm 0.18$), as a mechanism check rather than the main claim.
- Controlled AdamW reproduces the same trend, β_2 robustness, amplitude identity, and spectral-mixture behavior.
- A real $\sim 10\text{M}$ transformer shows no-fitting rate dependence: the fast sweep ends higher despite more cumulative LR.

Practical value. In data-poor sharp-decay prediction, low-calibration DropRelaxS transfer reduces held-out MAE by 44% on public curves (6/6 wins).

What remains open

Constants are measured, not solved.

- λ_{slow} is best interpreted as a slow preconditioned mode $\sim 2(\lambda/s)_{\text{eff}}$, not a universal spectral average.
- A robust first-principles predictor still needs an unexplained $O(1)$ constant.
- The normalization of LR drops must be stated explicitly; otherwise $c \approx 0.5$ and $c \approx 60\text{--}86$ can appear contradictory while describing the same data under different conventions.

Scope limits.

- Real-data public curves cover one architecture family and three scales.
- The $\sim 10\text{M}$ independently trained model confirms the effect but not a clean inverse-LR relaxation trend.
- The relationship to MPL's empirical exponent γ is complementary and still only qualitative.

Next work: highest-value experiments

Priority 0: strengthen current-law calibration.

- Add more two-stage probe LRs to estimate χ and λ_{slow} with tighter confidence intervals.
- Pre-register $\lambda = 10$ and the cross-scale c rule, then evaluate on new WSD shapes.
- Test whether the same c transfers across architectures or only within one family.

Priority 1: real-model amplitude identity.

- Train 3–4 constant-LR curves at $\sim 10\text{M}$.
- Independently estimate $dL_{\text{eq}}/d\eta$.
- Compare it against the DropRelaxS fitted amplitude.

Priority 2: S-time control.

- Alter batch size or effective LR integration so step-time and S -time are separated.
- Directly test whether memory follows $S_t - S_k$ rather than $t - k$.

Priority 3: scale and spectrum.

- If compute allows, repeat the real-transformer relaxation probe at $\geq 25\text{M}$.
- If diagnostics are available, measure Hessian/noise spectra and test the slow-mode explanation for λ_{slow} .

Reproducibility map

Result	Script / document
MPL reproduction and public-curve baseline	repro/reproduce_cosine_to_wsd.py
Residual structure and DropRelaxS regression	repro/nonadiabatic_theory.py
Relaxation-time check on public curves	repro/deep_tau.py
S-time vs step-time kernels	repro/deep_stime.py, deep_stime2.py
Spectral mixture diagnostics	repro/deep_rigor.py
Low-calibration cross-scale transfer	repro/deep_predict.py
Fair-baseline check	repro/correction_fair.py
Current-law calibration audit	repro/current_law_calibration_search.py
Current-law split research	repro/current_law_split_research.py
More-data calibration research	repro/current_law_more_data_calibration.py
Independent NQM simulations	represent/repro/E*.py, represent/repro/G*.py
Real ~ 10 M transformer training	represent/repro/train*.py
Independent audit and final report	represent/results/AUDIT_PARTC.md, represent/REPORT.md

Dependencies: `pip install -r requirements.txt`.

Main paper: `paper/main.tex`. Independent reproduction: `represent/REPORT.md`.

Code, running instructions, and work division

Repository.

<https://github.com/wjjpku/DL-final>

Minimal running instructions.

- 1 Install dependencies: `pip install -r requirements.txt`.
- 2 Reproduce main public-curve results: `python3 repro/reproduce_cosine_to_wsd.py -scales 25 100 400`.
- 3 Run key non-adiabatic diagnostics: `python3 repro/deep_tau.py`, `python3 repro/deep_predict.py`, `python3 repro/nonadiabatic_theory.py`.
- 4 Build paper and slides: `pdflatex paper/main.tex`; `pdflatex slides/main.tex`.

Work division. This is a single-member project. Jiaju Wu completed data processing, baseline reproduction, theory derivation, new correction design, experiments, independent reproduction analysis, writing, and slide preparation. Third-party code/data dependencies are kept under `external/` and cited in the paper.

Learning-rate schedules are not adiabatic.

Substantive conclusion. MPL/FSL-style laws capture strong quasi-static structure, but fast LR decay leaves a finite-time relaxation lag. The leading correction is a causal convolution of LR drops with a memory kernel in cumulative LR time.

Practical conclusion. DropRelaxS is useful as a low-calibration prior when full WSD curves are unavailable, especially for sharp-decay extrapolation.

Scientific boundary. The mechanism is real; the exact constants are not yet first-principles solved. The honest open problem is to compute λ_{slow} and c from spectra, and to test the relaxation trend on larger independently trained models.